The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 July 2009.

`INCH-POUND

MIL-PRF-19500/374E
30 April 2009
SUPERSEDING
MIL-PRF-19500/374D
29 June 1999

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPES 2N3996 THROUGH 2N3999,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power transistors for use in high-speed power switching applications. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for the unencapsulated die.

* 1.2 Physical dimensions. See figure 1 (types 2N3996 and 2N3997, 4 lead stud package), figure 2 (types 2N3998 and 2N3999, 3 lead stud package), figure 3, figure 4 (JANHC and JANKC, B-version), and figure 5 (JANHC and JANKC, C-version).

* 1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

P_T (1) T_A = +25°C	P_T (2) T_C = +100°C	V_{EBO}	V_{CBO}	V_{CEO}	I_B	I_C	I_C (3)	T_{STG} and T_J	$R_{\Theta JC}$
W	W	V dc	V dc	V dc	A dc	A dc	A dc	°C	°C/W
2	30	8	100	80	0.5	5.0	10	-65 to +200	3.33

(1) Derate linearly, 11.4 mW/°C for $T_A \geq$ +25°C.
(2) Derate linearly, 300 mW/°C for $T_C \geq$ +100°C.
(3) This value applies for $t_p \leq 1$ ms, duty cycle ≤ 50percent.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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1.4 Primary electrical characteristics at T_C = +25°C.

Limit	h_{FE2} (1) V_{CE} = 2 V dc; I_C = 1 A dc		$\|h_{fe}\|$ V_{CE} = 5 V dc; I_C = 1 A dc; f = 10 MHz	$V_{BE\ (sat)\ 2}$ (1) I_C = 5 A dc; I_B = 500 mA dc	$V_{CE\ (sat)\ 2}$ (1) I_C = 5 A dc; I_B = 500 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz
	2N3996 2N3998	2N3997 2N3999		V dc	V dc	pF
Minimum	40	80	3			
Maximum	120	240	12	1.6	2.0	150

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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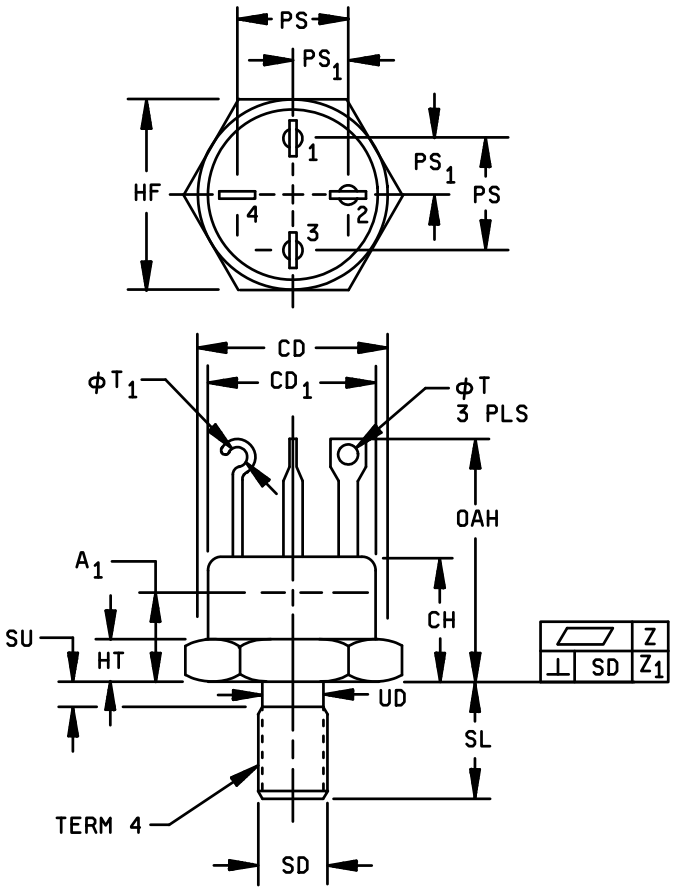


FIGURE 1. Physical dimensions for transistor types 2N3996 and 2N3997.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.345	.400	8.76	10.16	
A_1		.250		6.35	3
CD	.370	.437	9.40	11.10	3
CD_1	.318	.380	8.08	9.65	
HF	.424	.437	10.77	11.10	
PS	.180	.215	4.57	5.46	5
PS_1	.080	.110	2.03	2.79	5
HT	.090	.140	2.29	3.56	2,6
OAH	.575	.675	14.61	17.15	1
UD	.155	.189	3.94	4.80	
SL	.400	.455	10.16	11.56	
SU		.078		1.98	7
ϕT	.040	.065	1.02	1.65	
ϕT_1	.040	.065	1.02	1.65	4
SD	.190-32 UNF-2A				8
Z		.002		0.05	
Z_1		.006		0.15	

NOTES:

1. Terminal 1, emitter; terminal 2, base; terminal 3, collector; terminal 4, case.
2. Chamfer or undercut on one or both ends of hexagonal portion is optional.
3. The outline contour with the exception of the hexagon is optional within cylinder defined by CD_1 and A_1.
4. Terminal r can be flattened and pierced or hook type. A visual index is required when the flattened and pierced tab terminal contour (identical to the adjacent terminals) option is used. The case terminal (hook) is mechanically connected to the case. The other three terminals shall be electrically isolated from the case.
5. Angular orientation of terminals with respect to hexagon is optional.
6. HT dimension does not include sealing flanges.
7. SU is the length of incomplete or undercut threads.
8. SD is the pitch diameter of coated threads. Reference: Screw threads standards for Federal Service Handbook H28, part I.
9. Dimensions are in inches.
10. * Millimeters are giving for general information only.
11. * In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions for transistor types 2N3996 and 2N3997 - Continued.

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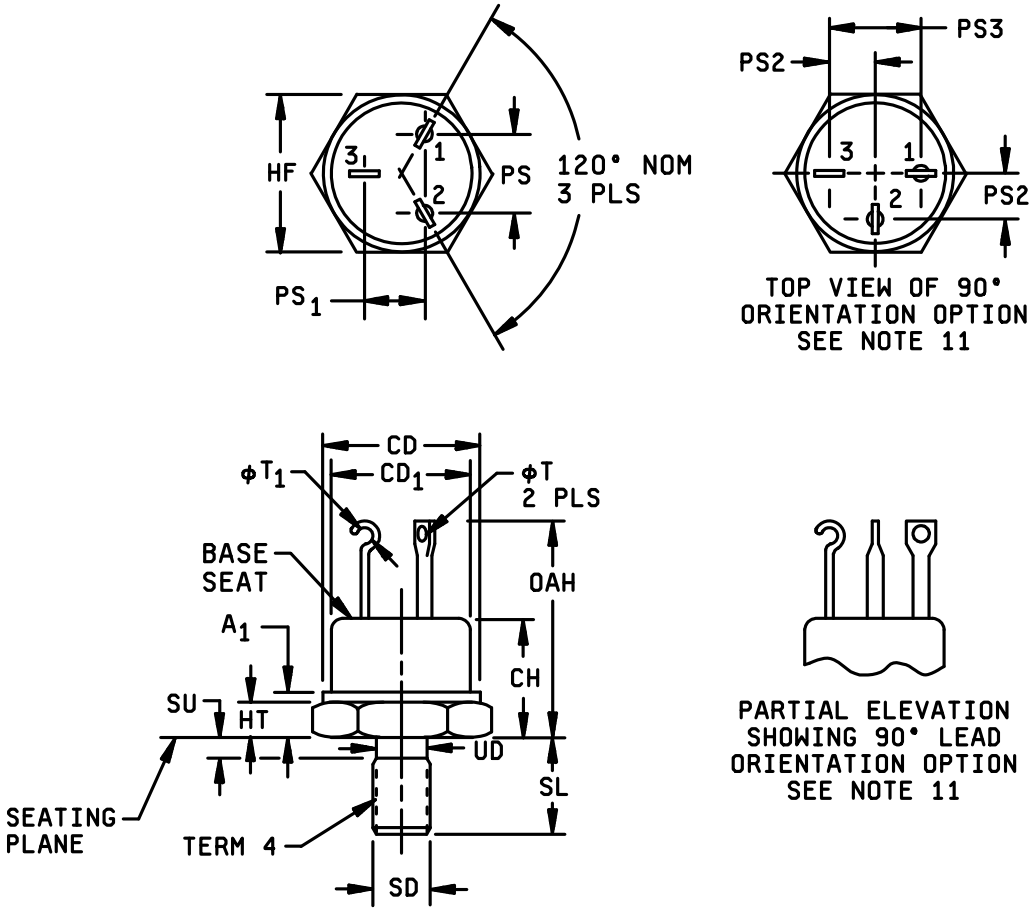


FIGURE 2. Physical dimensions for transistor types 2N3998 and 2N3999.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.345	.400	8.76	10.16	
A_1		.250		6.35	
CD_1	.318	.380	8.08	9.65	
CD	.370	.437	9.40	11.10	
HF	.424	.437	10.77	11.10	
PS	.125	.165	3.18	4.19	4,7,8
PS_1	.110	.145	2.79	3.68	4,7
PS_2	.090	.140	2.29	3.56	4,7,8
PS_3	.185	.215	4.70	5.46	4,7,8
HT	.090	.140	2.29	3.56	
OAH	.575	.675	14.61	17.15	5
UD	.155	.189	3.94	4.80	
SL	.400	.455	10.16	11.56	
SU		.078		1.98	9
ϕT	.040	.065	1.02	1.65	
ϕT_1	.040	.065	1.02	1.65	
SD	.190-32 UNF-2A				3

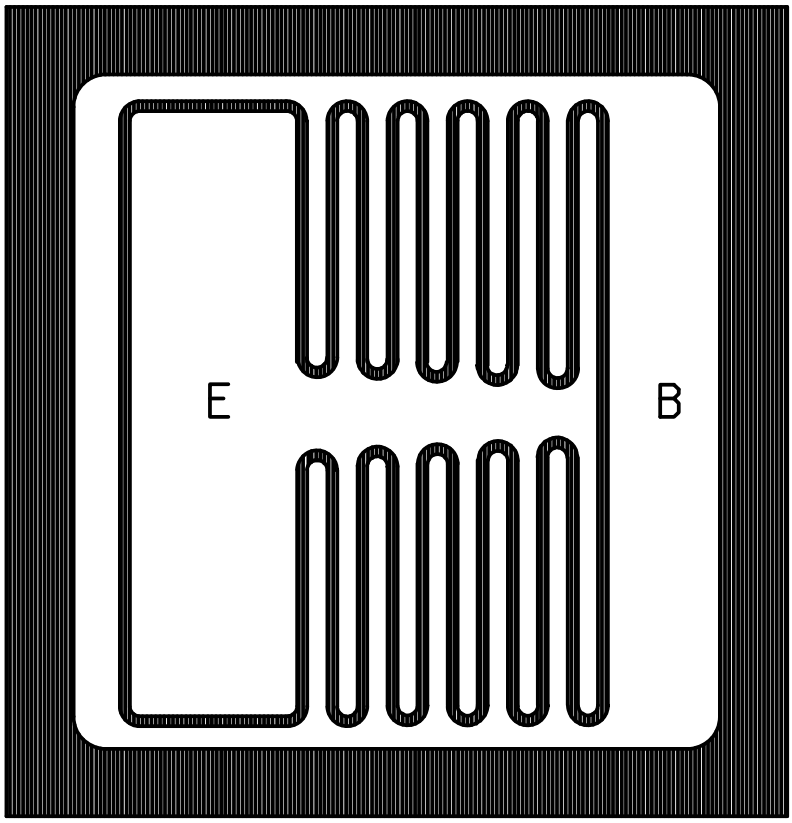
NOTES:

* 1. Dimensions are in inches. Millimeters are given for general information only.
2. Collector shall be electrically connected to the case. This terminal may be flattened and pierced only when the 90 degree option is used.
3. SD is the pitch diameter of coated threads. Reference: Screw thread standards for Federal Service Handbook H28, part I.
4. The orientation of the terminals in relation to the hex flats is not controlled.
5. All three terminals.
6. The case temperature may be measured anywhere on the seating plane within .125 (3.18 mm) of the stud.
7. Terminal spacing measured at the base seat only.
8. Dimensions PS, PS_1, PS_2, and PS_3 are measured from the centerline of terminals.
9. Maximum unthreaded dimension.
10. This dimension applies to the location of the center line of the terminals.
11. A 90 degree angle lead orientation as shown may be used at the option of the manufacturer. All dimensions of the basic outline except PS, PS_1, and the 120°lead angle apply to this option.
12. Terminal 1, emitter; terminal 2, base; terminal 3, collector.
13. A slight chamfer or undercut on one or both ends of the hexagonal is optional.

* 14. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 2. Physical dimensions for transistor types 2N3998 and 2N3999 - Continued.

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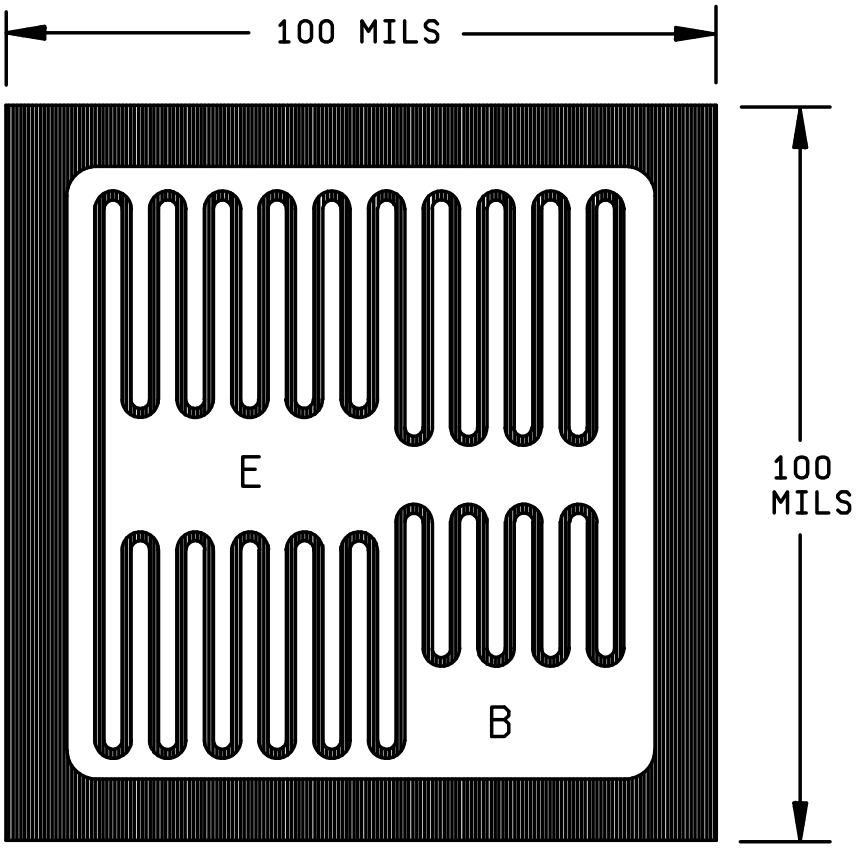


NOTES:

1. Chip size..... 82 X 82 mils
2. Chip thickness..... 6 to 12 mils
3. Top metal ..... Aluminum 25,000 Å minimum, 30,000 Å nominal
4. Back metal..... Gold 2,500 Å minimum, 3,000 Å nominal
5. Backside..... Collector
6. Bonding pad ..... B = 8 x 60 mils, E = 8 x 50 mils

FIGURE 3. JANHC and JANKC (A-versions) die dimensions.

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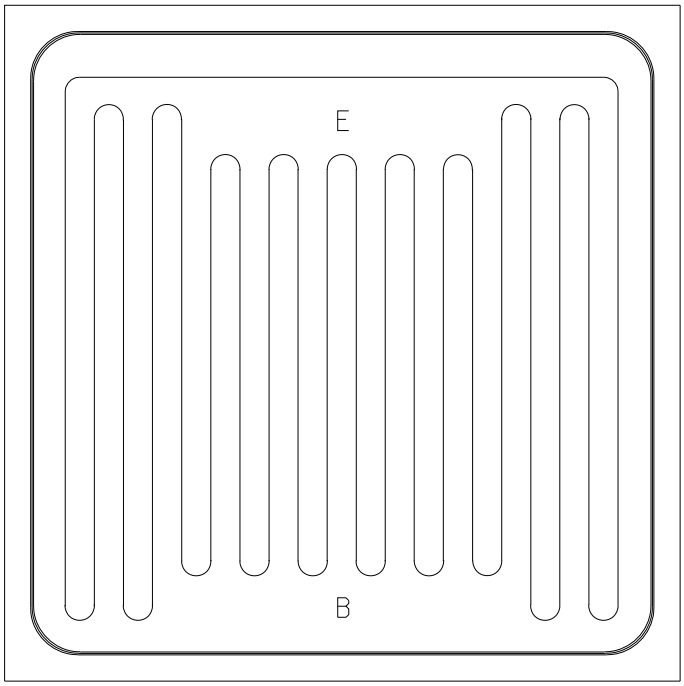


NOTES:

1. Chip size..... 100 X 100 mils
2. Chip thickness..... 6 to 12 mils
3. Top metal ..... Aluminum 25,000 Å minimum, 33,000 Å nominal
4. Back metal..... Gold 1,500 Å minimum, 2,500 Å nominal
5. Backside..... Collector
6. Bonding pad ..... 12 x 30 mils

FIGURE 4. JANHC and JANKC (B-versions) die dimensions.

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NOTES:

1.	Die size	.120 inch (3.05 mm) x .120 inch (3.05 mm) ±.002 inch (±0.05 mm).
2.	Die thickness	.014 inch (0.35 mm) ±.0015 inch nominal (±0.04 mm).
3.	Top metal	Aluminum, 54,000Å minimum, 60,000Å nominal.
4.	Back metal	Gold 6,000Å minimum, 8,000Å nominal.
5.	Backside	Collector
6.	Bonding pad	B = .060 x .012 inch (1.5 mm x 0.30 mm). E = .050 x .012 inch (1.27 mm x 0.30 mm).

* FIGURE 5. JANHC and JANKC (C-versions) die dimensions.

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

* 3.4 Interface and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1, 2, 3, 4, and 5 herein.

* 3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4 and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.8.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended.

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with appendix G of MIL-PRF-19500. This testing may be performed utilizing a TO-5 package in lieu of the TO-59 or the TO-111.

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* 4.3 Screening (JANS, JANTXV, JANTX levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{CES1} and h_{FE2}	I_{CES1}
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 100 nA dc, whichever is greater; Δh_{FE2} = +15 percent, -10 percent	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 100 nA dc; whichever is greater.
12	See 4.3.1	See 4.3.1
13a	Subgroups 2 and 3 of table I herein; ΔI_{CES1} = 100 percent of initial value or 100 nA dc, whichever is greater; Δh_{FE2} = +15 percent, -10 percent	Subgroup 2 of table I herein, ΔI_{CES1} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = +20 percent, -10 percent
13b	Insulation resistance (terminal to case) method 1016 of MIL-STD-750 (types 2N3996 and 2N3997 only); test condition B (short collector, emitter, and base terminals together); R_{iso} = 10^9 Ω minimum	Insulation resistance (terminal to case) method 1016 of MIL-STD-750 (types 2N3996 and 2N3997 only); test condition B (short collector, emitter, and base terminals together); R_{iso} = 10^9 Ω minimum.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions for all levels are as follows: V_{CE} = 25 V dc, ±5 V dc; T_J = 187.5°C, ±12.5°C; T_A = 35°C. max.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with appendix G of MIL-PRF-19500. Test limits and conditions shall be chosen by the supplier to demonstrate compliance with electrical characteristics specified in table I herein.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF19500 and as specified herein. Group A inspection shall be performed on each sublot.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIA (JANS) and E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	$V_{CE} \geq 10$ V dc, 2,000 cycles.
B5	1027	V_{CE} = 20 V dc ±1.0 V dc; 96 hours. Adjusted as required by the chosen T_A to give an average lot T_J = +275°C.
B7	3053	Load condition C (unclamped inductive) (see figure 6), T_A = +25°C ±3°C, duty cycle ≤ 10 percent; $R_S = 0.1\Omega$. Test 1 - t_p = 640 µs; $R_{BB1} = 39\Omega$; V_{BB1} = 20 V dc; $R_{BB2} = \infty$; V_{BB2} = 0; V_{CC} = 10 V dc; I_C = 4.3 A dc; L = 1 mH (0.5Ω, 5A) (Tower #7870 or equivalent). Test 2 - t_p = 2.88 mS; $R_{BB1} = 120\Omega$; V_{BB1} = 20 V dc; $R_{BB2} = \infty$; V_{BB2} = 0; V_{CC} = 10 V dc; I_C = 1.4 A dc; L = 10 mH (0.11 Ω, 12.5A) (Stancor C-2688 or equivalent).
B7	3053	Safe operating area (switching) (destructive); load condition B (clamped inductive) T_A = +25°C; I_B = 0.5 A dc, I_C = 5 A dc; V_{CC} = 55 V dc, L = 1.0 mH, V_{BB2} = 3.0 V, R_{BB2} = 20 Ω.

4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	For solder die attach; $V_{CE} \geq 10$ V dc, 2,000 cycles.
B3	1027	For eutectic die attach T_A = +35°C max, P_T adjusted to achieve T_J = +175°C min, $V_{CE} \geq 10$ V dc.
B5		Not applicable.
B7	3053	Load condition C (unclamped inductive) (see figure 6), T_A = +25°C, duty cycle ≤ 10 percent; R_S = 0.1 Ω. Test 1 - t_p = 640 µs; $R_{BB1} = 39\Omega$; V_{BB1} = 20 V dc; $R_{BB2} = \infty$; V_{BB2} = 0; V_{CC} = 10 V dc; I_C = 4.3 A dc; L = 1 mH (0.5 Ω, 5A) (Tower #7870 or equivalent). Test 2 - t_p = 2.88 mS; $R_{BB1} = 120\Omega$; V_{BB1} = 20 V dc; $R_{BB2} = \infty$; V_{BB2} = 0; V_{CC} = 10 V dc; I_C = 1.4 A dc; L = 10 mH (0.11 Ω, 12.5A) (Stancor C-2688 or equivalent)
B7	3053	Safe operating area (switching) (destructive); load condition B (clamped inductive) T_A = 25°C; I_B = 0.5 A dc, I_C = 5 A dc; V_{CC} = 55 V dc, L = 1.0 mH, V_{BB2} = 3.0 V, R_{BB2} = 20 Ω.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Test condition A; weight = 7 pounds ±5 ounce, application time = 15 s, tubulated leads only. Test condition D2; torque = 15 inch-pound; application time = 15 s. Test condition D1; torque = 8 in-oz; application time = 15 s; tubulated leads only.
* C5	3131	See 4.5.4, $R_{\theta JC}$ = 5°C/W.
C6	1037	For solder die attach; $V_{CE} \geq 10$ V dc, 6,000 cycles
C6	1026	For eutectic die attach T_A = +35°C max, P_T adjusted to achieve T_J = +175°C min, $V_{CE} \geq 10$ V dc

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Case-temperature control for h_{fe} test. To maintain the case temperature at less than +40°C for this test, the specified dc collector current should be applied for not longer than 10 seconds without employing a heat sink.

4.5.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted as a case temperature (T_C) of +25°C.

* 4.5.4 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, and t_H. Measurement delay time t_{MD} = 70 ms max. Forced moving air or draft shall not be permitted across the devices during test.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
* Thermal impedance 2/	3131	See 4.3.3	$Z_{\theta JX}$			°C /W
Breakdown voltage, collector to base	3001	Bias condition D, I_C = 10 μA dc, I_B = 0	$V_{(BR)CBO}$	100		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 50 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 80 V dc, V_{BE} = 0	I_{CES1}		200	nA dc
Collector to emitter cutoff current	3041	Bias condition D, V_{CE} = 60 V dc, I_B = 0	I_{CEO}		10	μA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5 V dc, I_C = 0	I_{EBO1}		200	nA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 8 V dc, I_C = 0	I_{EBO2}		10	μA dc
Base-emitter voltage (saturated)	3066	Test condition A, I_C = 1 A dc, I_B = 0.1 A dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.2	V dc
Base-emitter voltage (saturated)	3066	Test condition A, I_C = 5 A dc, I_B = 0.5 A dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.6	V dc
Collector to emitter voltage (saturated)	3071	I_C = 1 A dc; I_B = 0.1 A dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.25	V dc
Collector to emitter voltage (saturated)	3071	I_C = 5 A dc; I_B = 0.5 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		2	V dc

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE1}			
2N3996, 2N3998				30		
2N3997, 2N3999				60		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N3996, 2N3998				40	120	
2N3997, 2N3999				80	240	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5 A dc, pulsed (see 4.5.1)	h_{FE3}			
2N3996, 2N3998				15		
2N3997, 2N3999				20		
Subgroup 3						
High temperature operation		T_C = +150°C				
Collector to emitter cutoff current	3041	Bias Condition C, V_{CE} = 80 V dc, V_{BE} = 0	I_{CES2}		50	µA dc
Low-temperature operation		T_C = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE4}			
2N3996, 2N3998				10		
2N3997, 2N3999				20		

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		150	pF
Magnitude of common emitter small-signal short circuit forward current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 A dc; f = 10 MHz, (see 4.5.2)	h_{fe}	3	12	
Switching parameters:						
2N3996 and 2N3998						
Pulse delay time		See figure 7	t_d		100	ns
Pulse rise time		See figure 7	t_r		240	ns
Pulse storage time		See figure 7	t_s		1.4	µs
Pulse fall time		See figure 7	t_f		.3	µs
t_{on}		$t_d + t_r$	t_{on}		.3	µs
t_{off}		$t_s + t_f$	t_{off}		1.5	µs
2N3997 and 2N3999						
Pulse delay time		See figure 7	t_d		100	ns
Pulse rise time		See figure 7	t_r		240	ns
Pulse storage time		See figure 7	t_s		1.75	µs
Pulse fall time		See figure 7	t_f		.3	µs
t_{on}		$t_d + t_r$	t_{on}		.3	µs
t_{off}		$t_s + t_f$	t_{off}		2.0	µs
Subgroup 5						
Safe operating area (DC)	3051	T_C = 100°C; power application time = 1.0 sec. (see figure 8)				
Test 1		V_{CE} = 80 V dc; I_C = .080 A dc				
Test 2		V_{CE} = 20 V dc; I_C = 1.5 A dc				

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - continued						
Safe operating area (clamped inductive)		(See figures 9 and 10); (clamped inductive load) T_A = 25°C; I_B = 0.5 A dc; I_C = 5 A dc; V_{CC} = 15 V dc; Load condition C				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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* TABLE II. Groups A, B, C, and E delta measurements. 1/ 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1 A dc pulsed (see 4.5.1)	Δh_{FE2}		±20 percent change from initial reading.	
2.	Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 60 V dc I_B = 0	ΔI_{CEO}		100 percent of initial value or 100 nA dc, whichever is greater.	
3.	Collector to emitter voltage (saturated)	3071	I_C = 1 A dc I_B = 0.1 A dc pulsed (see 4.5.1)	$\Delta V_{CE(SAT)1}$		±50 mV dc change from previously measured value.	
4.	Thermal impedance	3131	See 4.3.3	$\Delta Z_{\theta JX}$		3.3	°C/W

1/ The devices which exceed the group A limits for this test shall not be acceptable.
2/ The delta measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroups 4 and 5, see table II herein, steps 1 and 4.
   b. Subgroup 7, see table II herein, step 1.
3/ The delta measurements for table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, steps 1 and 4.
   b. Subgroup 6, see table II herein, step 2.
   c. Subgroup 7, see table II herein, step 1.
4/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, step 3 for JANS.
   b. Subgroup 6, see table II herein, steps 1, 2, and 4 for JANS and steps 1 and 4 for JAN, JANTX, and JANTXV.
5/ The delta measurements for table E-IX of MIL-PRF-19500 are: Subgroup 1, see table II herein, all steps.

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* TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	500 cycles minimum	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		Table I, subgroup 2 and table II herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C, V_{CB} = 80 percent of rated, T = 1,000 hours.	
Electrical measurements		Table I, subgroup 2.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 8			22 devices c = 0
Reverse stability	1033	Condition B	

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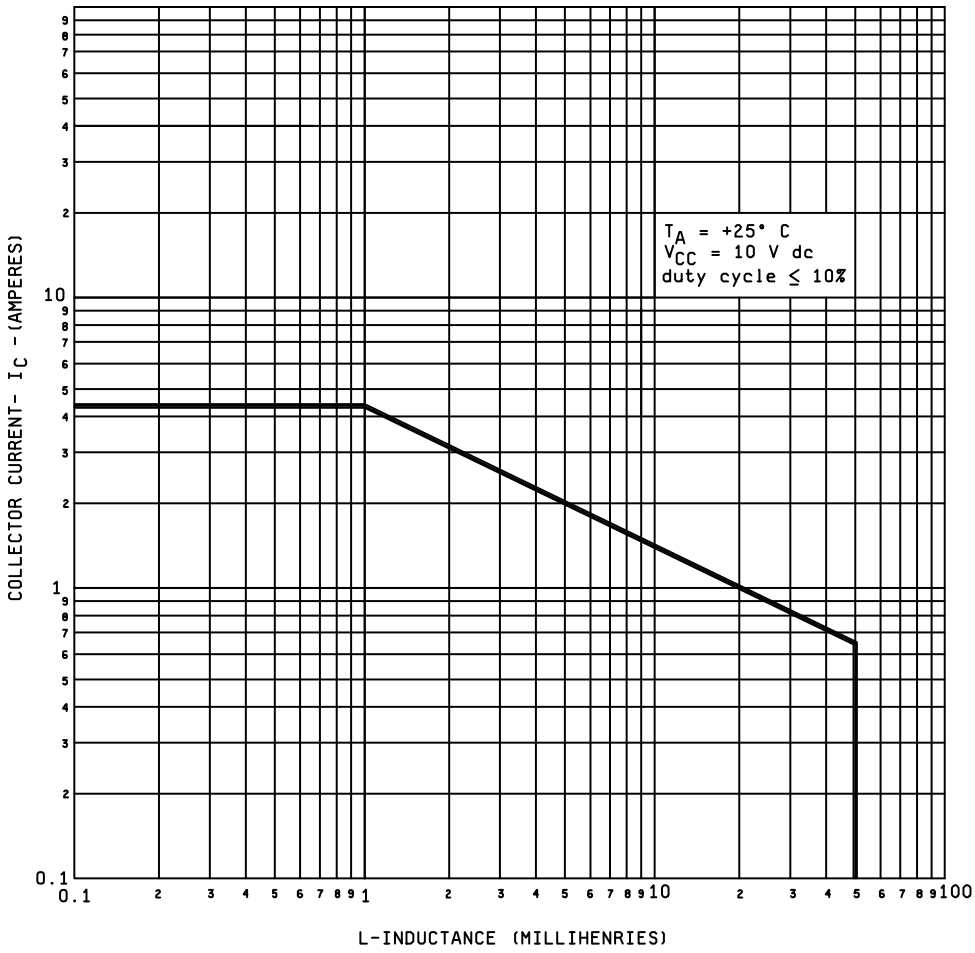
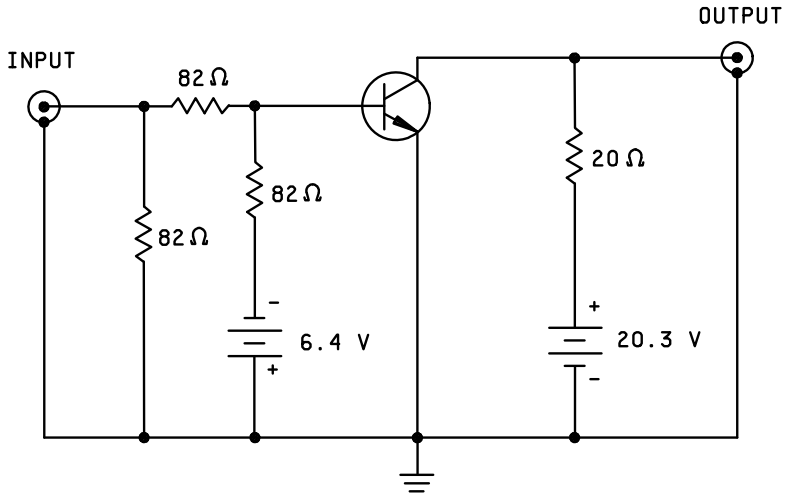
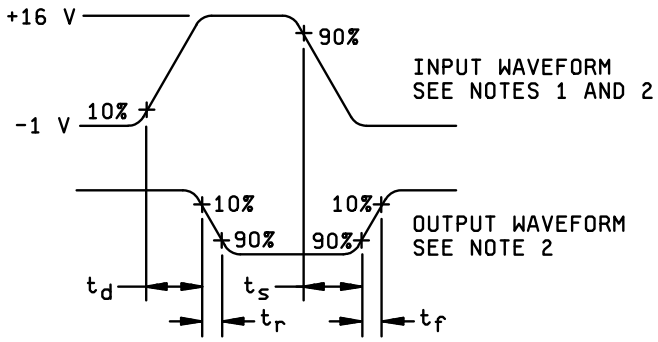


FIGURE 6. Safe operating area for switching between saturation and cutoff - unclamped inductive load.

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NOTES:

1. The input waveform is supplied by a generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{out} = 50\ \Omega$, pW = 2 μs, duty cycle ≤ 2 percent.
2. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $R_{in} \geq 10\ M\Omega$, $C_{in} \leq 11.5$ pF.
3. Resistors must be non-inductive types.
4. The d-c power supplies may require additional by-passing in order to minimize ringing.

FIGURE 7. Pulse response test circuit.

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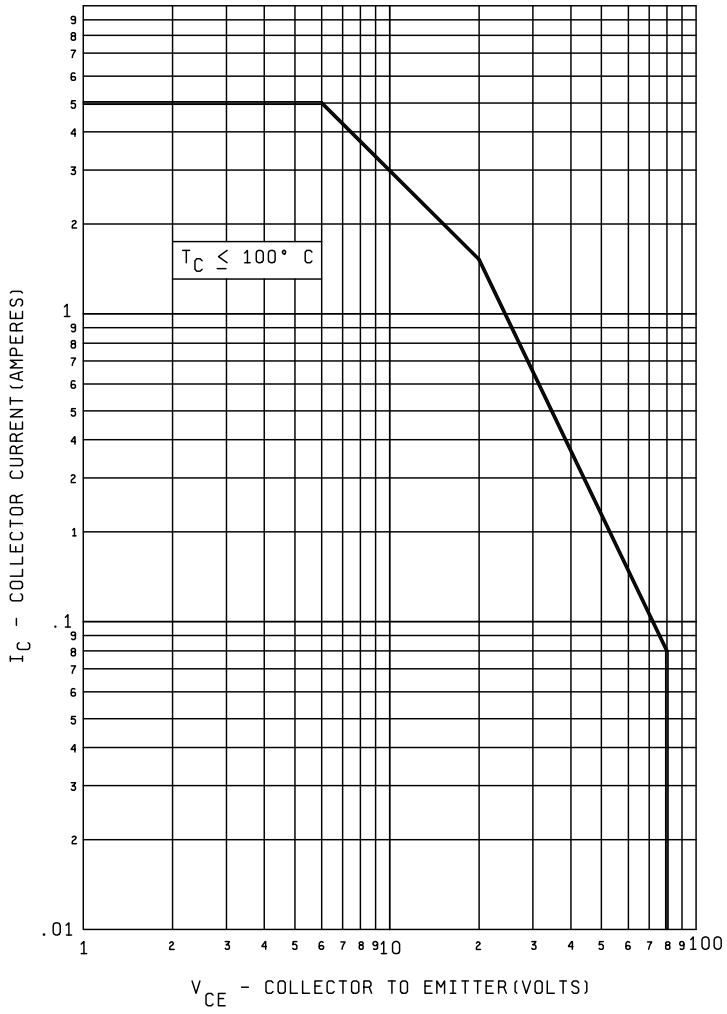
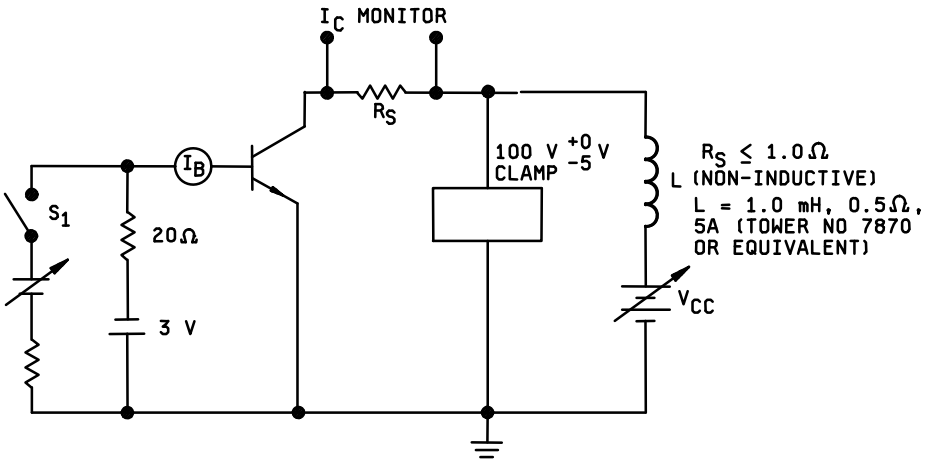


FIGURE 8. Maximum safe operating area graph (dc).

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Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached.
3. Perform specified end point test.

FIGURE 9. Clamped inductive sweep test circuit.

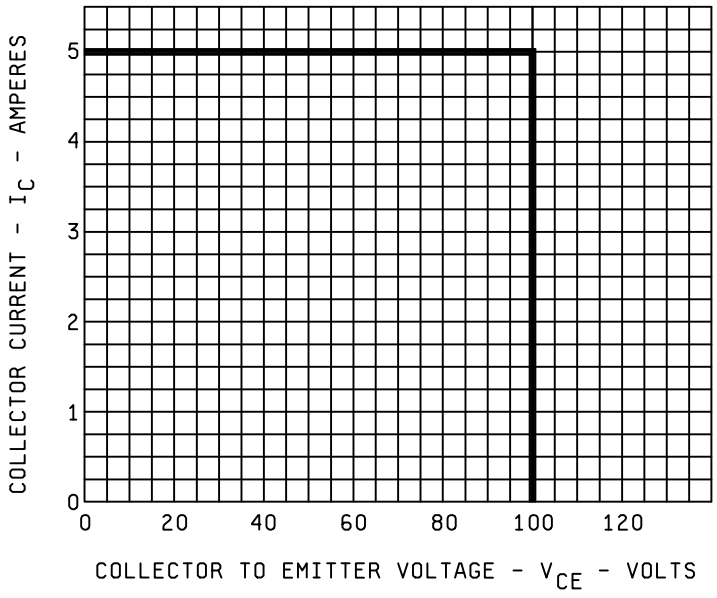


FIGURE 10. Safe operating area for switching between saturation and cutoff - clamped inductive load.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

* 6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCB2N3996) will be identified on the QML.

JANHC and JANKC ordering information			
Pin	Manufacturer		
	33178 (1)	34156	43611
2N3996	JANHCA2N3996	JANHCB2N3996 JANKCB2N3996	JANHCC2N3996
2N3997	JANHCA2N3997	JANHCB2N3997 JANKCB2N3997	JANHCC2N3997
2N3998	JANHCA2N3998	JANHCB2N3998 JANKCB2N3998	JANHCC2N3998
2N3999	JANHCA2N3999	JANHCB2N3999 JANKCB2N3999	JANHCC2N3999

(1) This die version is no longer manufactured.

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AV
Air Force - 99

Preparing activity:
DLA - CC

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